



CLOUD & DEVOPS ENGINEERING

Technical Assessment 2025

Objective: Evaluate deep operational understanding of Cloud infrastructure design, modern orchestration pipelines, high-availability architecture, network topology security, and continuous delivery mechanisms.

Project Brief

Design, build, secure, and deploy the entire server, application, and database architecture for CURE's distributed digital health environment over AWS cloud infrastructure.

Core Infrastructure Requirements

- **High Availability Cloud Deployment:** Provision and configure virtual computing instances, secure object storage, and highly isolated relational databases with multi-AZ behaviors.
- **CI/CD Pipelines:** Establish fully automated deployment configurations triggered upon mainline Git branch approvals, ensuring automated testing and container assembly.
- **Network Isolation & Traffic Routing:** Enforce SSL/TLS encryption across all endpoints. Set up edge reverse proxies and CDNs to ensure fast content delivery.

Expected AWS Cloud Architecture

AWS EC2

AWS S3

AWS RDS

AWS CloudFront

AWS Route53

AWS IAM

Enterprise / Bonus Requirements

IaC via Terraform

Container Orchestration (EKS)

Prometheus + Grafana



CORPORATE EVALUATION PHILOSOPHY

1. ENGINEERING MINDSET

Your technical choices must be deliberate and justified. Design architectures to treat components as interchangeable blocks, backed by comprehensive documentation.

2. PRODUCTION READINESS

We evaluate code based on fitness for production. This requires thorough input validation, robust exception handling, and structural security.

3. ATTENTION TO DETAIL

This includes clean variable naming conventions, consistent formatting, comprehensive error messaging, and intuitive edge-case handling.

4. CODE DELIVERY PROTOCOL

We look for professional utilization of version control, semantic commit structures, and a polished presentation formatting layout for all deliverables.



Service Level Agreement (SLA)

All core code, architecture blueprints, live links, and descriptive video demonstrations must be submitted within **72 Hours** from assignment delivery.